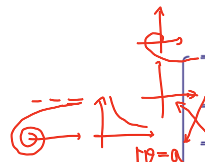
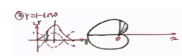
高等数学

- 套公式
- ☆画图
- ☆计算

- 几何应用
  - 研究对象
    - f(x), fn(x), 参数方程，【偏导数】（和13讲结合），原函数，【微分方程的解函数f(x)】
    - 函数f(x)
    - 函数列fn(x)
    - 参数方程
    - 偏导数(f(x,y)=0 -> y=y(x))
      12'
    - 变限积分
    - 微分方程的解函数f(x)
      12'
  - 研究内容
    5' 12'
    - 面积
      - 直角坐标系 $S=\int_a^b|f(x)-g(x)|dx$
        - 常见图
        - 非常见图
          - 描特殊点
          - 用图像变换
          - 导数工具
            - 用极限
            - 一阶导数定单调
            - 二阶导数定凹凸
      - 极坐标系 $S=\int_\alpha^\beta\frac{1}{2}|r_2^2(\theta)-r_1^2(\theta)|d\theta$
        - 常见图
          - 阿基米德螺线 $r=\theta$
          - 对数螺线 $r=\frac{1}{\theta}$
          - 双曲螺线 $r=e^\theta$
          - 概要：$S=\int_a^b\frac{1}{2}r^2d\theta$



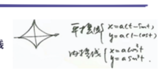
          - 心形线
            - 

              r=1+cosø
            - 

              r=1-cosø
            - 

              r=1+sinø
            - 

              r=1-sinø
        - 非常见图
          - 描点法
          - 用直角系观点画极坐标系的图
      - 参数法
        - 常见图
          - 外摆线
          - 平摆线 

          - 内摆线 

        - 非常见法
          - 换元法！$\int_a^b f(x)dx=\int_\alpha^\beta y(t)x'(t)dt$
    - 体积
      - 旋转体体积
        - ∫s 绕x轴 $V_x=\int_a^b \pi y^2(x)dx$
        - ∫ch 绕y轴 $V_y=\int_a^b 2\pi x|y(x)|dx$
        - ①套公式，考计算
        - ②考综合
        - ③考新颖
    - 平均值
    - 弧长
      - 直角坐标系 $s=\int_a^b\sqrt{1+[y'(x)]^2}dx$
      - 参数方程 $s=\int_\alpha^\beta\sqrt{[x'(t)]^2+[y'(t)]^2}dt$
      - 极坐标系 $s=\int_\alpha^\beta\sqrt{[r(\theta)]^2+[r'(\theta)]^2}d\theta$
      - 注意求弧长是小到大积分，换元法不要求
    - 侧面积
      - 直角坐标 $S=2\pi\int_\alpha^\beta|y(x)|\sqrt{1+[y'(x)]^2}dx=2\pi\int_\alpha^\beta|y(x)|ds$
      - 参数方程 $S=2\pi\int_\alpha^\beta|y(t)|\sqrt{[x'(t)]^2+[y'(t)]^2}dt=2\pi\int_\alpha^\beta|y(t)|ds$
      - 不用dx而用ds
    - 形心
      - $\overline{x}_{简单}=\frac{\iint_D xd\sigma}{\iint_D d\sigma}=\frac{\int_a^b dx\int_0^{f(x)}xdy}{\int_a^b dx\int_0^{f(x)}dy}=【\frac{\int_a^b f(x)xdx}{\int_a^b f(x)dx}】$
      - $\overline{y}_{简单}=\frac{\iint_D yd\sigma}{\iint_D d\sigma}=\frac{\int_a^b dx\int_0^{f(y)}ydy}{\int_a^b dx\int_0^{f(x)}dy}=【\frac{\frac{1}{2}\int_a^b f^2(x)dx}{\int_a^b f(x)dx}】$
      - $\overline{x}_{普通}=【\frac{\int_a^b x[f(x)-g(x)]dx}{\int_a^b[f(x)-g(x)]dx}】=\frac{\iint_D xd\sigma}{\iint_D 1d\sigma}$
      - $\overline{y}_{普通}=【\frac{\frac{1}{2}\int_a^b[f^2(x)-g^2(x)]dx}{\int_a^b[f(x)-g(x)]dx}】=\frac{\iint_D yd\sigma}{\iint 1d\sigma}$

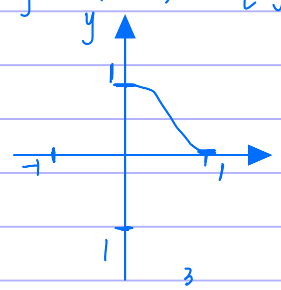
- 要求
  - 1. 会画图
  - 2. 套公式
  - 3. 做计算

Part One 课中例题

例10.3

例 10.3 求曲线 $y^2=(1-x^2)^3$ 所围图形的面积.

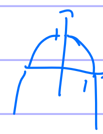
$y=\pm(1-x^2)^{\frac{3}{2}}$ [y轴对称，x轴对称]



$y'=\frac{3}{2}(1-x^2)^{\frac{1}{2}}\cdot(-2x)<0$

$y'(0)=0$

$1-x^2=$



$y=(1-x^2)^{\frac{3}{2}}$

$y'=\frac{3}{2}(1-x^2)^{\frac{1}{2}}(-2x)=-3x(1-x^2)^{\frac{1}{2}}$

$y''=-3(1-x^2)^{\frac{1}{2}}+3x^2\cdot(1-x^2)^{-\frac{1}{2}}$

$4\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}=\frac{3}{4}\pi$

$$S=4\int_0^1(1-x^2)^{\frac{3}{2}}dx\overset{x=\sin t}{=}4\int_0^{\frac{\pi}{2}}\cos^3t\,\underbrace{dt}_{\cos t\,dt}=\cancel{4\cdot\frac{2}{3}\cdot1=\frac{8}{3}}$$

例10.7

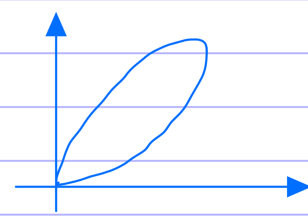
例 10.7 求曲线 $x^3+y^3-3axy=0(a>0)$ 所围成的平面图形的面积.

$x^3+y^3-3axy=0$

$r^3(\cos^3\theta+\sin^3\theta)=r^2\cdot3a\cdot\cos\theta\cdot\sin\theta$

$r(\cos^3\theta+\sin^3\theta)=3a\cdot\cos\theta\cdot\sin\theta$

$r=\frac{3a\cdot\cos\theta\cdot\sin\theta}{\cos^3\theta+\sin^3\theta}\qquad 0\le\theta\le\frac{\pi}{2}$



$$S=\int_0^{\frac{\pi}{2}}\frac{1}{2}r^2d\theta=\int_0^{\frac{\pi}{2}}\frac{9a^2}{2}\frac{\cos^2\theta\sin^2\theta}{(\cos^3\theta+\sin^3\theta)^2}d\theta$$

$$=\frac{9a^2}{2}\int_0^{\frac{\pi}{2}}\frac{\tan^2\theta\sec^2\theta\,d\theta}{(1+\tan^3\theta)^2}=\frac{9a^2}{2}\int_0^{\frac{\pi}{2}}\frac{\tan^2\theta\,d\tan\theta}{(1+\tan^3\theta)^2}=\frac{3a^2}{2}\int_0^{+\infty}\frac{d(1+x^3)}{(1+x^3)^2}=\frac{3a^2}{2}\left(-\frac{1}{1+\tan^3\theta}\right)\Big|_0^{\frac{\pi}{2}}=\frac{3}{2}a^2$$

例9.11

例 10.11 求曲线 $y=x\sqrt{4x-x^2}$ 在[0,4]上与 x 轴所围图形绕 y 轴旋转一周所得的旋转体的体积.

$\frac{1-\cos4u}{8}$

$2\pi r\cdot h$

$$V=\int_0^4 2\pi x\cdot x\sqrt{4x-x^2}\,dx\overset{\frac{t}{2}=\sin u}{=}8\pi\int_0^{\frac{\pi}{2}}4(1+\sin^2u)\cos^2u\cdot2\,du$$

$$=\int_0^4 2\pi x^2\sqrt{-(x-2)^2+4}\,dx=64\pi\int_0^{\frac{\pi}{2}}\cos^2u+\frac{\sin^22u}{4}\,du$$

$$\overset{x-2=t}{=}\int_{-2}^{2}2\pi(t+2)^2\sqrt{4-t^2}\,dt=64\pi\left[\frac{1}{2}\cdot\frac{\pi}{2}+\frac{1-\cos4\pi}{8}\Big|_0^{\frac{\pi}{2}}\right]$$

$$=8\pi\int_0^2(t+2)^2\sqrt{1-\left(\frac{t}{2}\right)^2}\,dt=64\pi\left(\frac{\pi}{4}+\frac{\pi}{16}\right)=20\pi$$

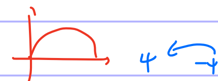
综合

例 10.14

例 10.14 设函数 $y=f(x)$ 满足微分方程 $y'+y=\dfrac{e^{-x}\cos x}{2\sqrt{\sin x}}$，且 $f(\pi)=0$，则曲线 $y=f(x)(x\geqslant 0)$ 绕 x 轴旋转一周所成旋转体的体积是（　　）.

(A) $\dfrac{\pi}{5(1+e^{-2\pi})}$　(B) $\dfrac{\pi}{5(1-e^{-2\pi})}$　(C) $\dfrac{\pi}{5(1+e^{-\pi})}$　(D) $\dfrac{\pi}{5(1-e^{-\pi})}$

$$y=e^{-\int P\,dx}\left(\int e^{\int P dx}\cdot q\,dx+C\right)$$
$$=e^{-x}\left(\int \frac{\cos x}{2\sqrt{\sin x}}dx+C\right)$$
$$=e^{-x}\left(\int\frac{d\sin x}{2\sqrt{\sin x}}+C\right)=e^{-x}\left(\sqrt{\sin x}+C\right)$$
$$f(\pi)=e^{-\pi}(0+C)=0\Rightarrow C=0$$
$$\Rightarrow f(x)=e^{-x}\sqrt{\sin x}\qquad \sin x\geqslant 0$$



$$V=\sum_{n=0}^{\infty}\int_{2n\pi}^{(2n+1)\pi}\pi e^{-2x}\sin x\,dx$$
$$=\sum_{n=0}^{\infty}\pi\int_{2n\pi}^{(2n+1)\pi}e^{-2x}\sin x\,dx$$

$$\begin{array}{ccc} e^{-2x} & -2e^{-2x} & 4e^{-2x}\\ \sin x & -\cos x & -\sin x\end{array}\qquad \frac{-e^{-2x}(\cos x+\sin x)}{5}$$

$$=\sum_{n=0}^{\infty}\pi\left.\frac{-e^{-2x}(\cos x+\sin x)}{5}\right|_{2n\pi}^{(2n+1)\pi}=\sum_{n=0}^{\infty}\pi\frac{e^{-2\pi(2n+1)}+e^{-2\pi(2n)}}{5}$$

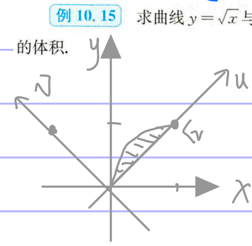
$$e^{-2\pi(2n+1)}+e^{-2\pi(2n)}$$

$$=\frac{\pi}{5}\sum_{n=0}^{\infty}e^{-4n\pi}\cdot e^{-2\pi}+e^{-4n\pi}=\frac{\pi(e^{-2\pi}+1)}{5}\sum_{n=0}^{\infty}e^{-4n\pi}=\frac{\pi(e^{-2\pi}+1)}{5}\cdot\frac{1}{1-e^{-4n\pi}}=\frac{\pi(e^{-2\pi}+1)}{5(1-e^{-4\pi})}$$

$\Rightarrow B$

例 10.15　肖

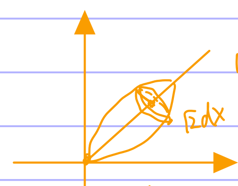
例 10.15 求曲线 $y=\sqrt{x}$ 与 $y=x$ 所围平面有界区域绕直线 $y=x$ 旋转一周所得旋转体的体积.

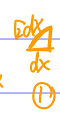


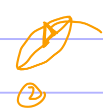
$$\begin{cases}u=\frac{\sqrt2}{2}x+\frac{\sqrt2}{2}y\\ v=-\frac{\sqrt2}{2}x+\frac{\sqrt2}{2}y\end{cases}\qquad\begin{cases}y=\frac{u+v}{\sqrt2}\\ x=\frac{u-v}{\sqrt2}\end{cases}$$

$$\Rightarrow y=\sqrt{x}\qquad \frac{u+v}{\sqrt2}=\sqrt{\frac{u-v}{\sqrt2}}\qquad \frac{(u+v)^2}{\sqrt2}=u-v$$

$$\int_0^{\sqrt2}\pi v^2\,du$$







$\frac{\sqrt{x}-x}{\sqrt2}$

$$V=\int_0^1\pi\left(\frac{\sqrt{x}-x}{\sqrt2}\right)^2\cdot(\sqrt2\,dx)$$

例 10.16

例 10.16 已知函数 $f(x)$ 在 $\left[0,\frac{3\pi}{2}\right]$ 上连续，在 $\left(0,\frac{3\pi}{2}\right)$ 内是函数 $\frac{\cos x}{2x-3\pi}$ 的一个原函数，且 $f(0)=0$.

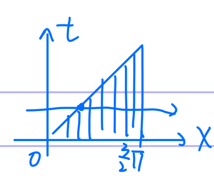
(1) 求 $f(x)$ 在区间 $\left[0,\frac{3\pi}{2}\right]$ 上的平均值；

(2) 证明 $f(x)$ 在区间 $\left(0,\frac{3\pi}{2}\right)$ 内存在唯一零点.

(1)

$$f(x)=f(0)+\int_0^x\frac{\cos t}{2t-3\pi}dt$$

$$\bar f=\frac{\int_0^{\frac32\pi}\left(\int_0^x\frac{\cos t}{2t-3\pi}dt\right)dx}{\frac32\pi}$$



$$= \frac{2}{3\pi}\int_0^{\frac{3}{2}\pi} dt \int_t^{\frac{3}{2}\pi} \frac{\cos t}{2t-3\pi} dx$$

$$\sin t \Big|_0^{\frac{3}{2}\pi} = -1$$

$$= \frac{2}{3\pi}\int_0^{\frac{3}{2}\pi} \frac{\cos t(\frac{3}{2}\pi - t)}{2t-3\pi} dt = \frac{-1}{3\pi}\int_0^{\frac{3}{2}\pi} \cos t\, dt = +\frac{1}{3\pi}$$

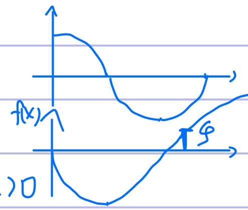
(2) $f(x) = \int_0^x \frac{\cos x}{2x-3\pi} dx$

$f'(x) = \frac{\cos x}{2x-3\pi}$

$x\in(0,\frac{\pi}{2}) \downarrow \quad f(\frac{\pi}{2})<0$

$x\in(\frac{\pi}{2},\frac{3}{2}\pi) \uparrow \quad f(\xi) = \frac{1}{3\pi}\cdot\frac{3}{2} = \frac{1}{2\pi} > 0$

$\xi\in(0,\frac{3}{2}\pi) \Rightarrow \xi\in(\frac{\pi}{2},\frac{3}{2}\pi)$



例 10.17

例 10.17 曲线 $y=\int_0^x \tan t\,dt\left(0\leqslant x\leqslant \frac{\pi}{4}\right)$ 的弧长 $s=$ ______.

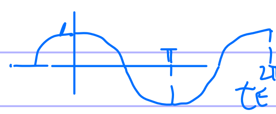
$$S = \int_0^{\frac{\pi}{4}} \sqrt{1+(y'_x)^2}\, dx$$

$$= \int_0^{\frac{\pi}{4}} \sqrt{1+\tan^2 x}\, dx$$

$$= \ln|\sec x + \tan x|\Big|_0^{\frac{\pi}{4}} = \ln(\sqrt{2}+1)$$

例 10.18

例 10.18 求曲线 $y=\int_{-\frac{\pi}{2}}^{x} \sqrt{\cos t}\,dt$ 的全长.



$t\in(-\frac{\pi}{2},\frac{\pi}{2})$

$$\int_{-\pi/2}^{\pi/2} \sqrt{1+\cos x}\, dx \qquad \sqrt{1-t}$$

$$= 2\int_0^{\pi/2} \sqrt{1+\sin x}\, dx$$

$$\overset{\sin x = t}{=} 2\int_0^1 \frac{\sqrt{1+t}}{\sqrt{1-t^2}} dt = 4\int_0^1 \frac{1}{2\sqrt{1-t}} d(1-t) = -4\sqrt{1-t}\Big|_0^1 = 4$$

例 10.19

例 10.19 当 $x\geqslant 0$ 时,曲线 $y=\frac{1}{4}\int_0^2 x\sqrt{12-x^2t^2}\,dt$ 的全长为______.

$$y = \frac{x}{4}\int_0^2 \sqrt{12-x^2t^2}\, dt$$

$xt = u$, $t: 0 \to 2$, $u: 0 \to 2x$

$$= \frac{1}{4}\int_0^{2x} \sqrt{12-u^2}\, du$$

$12-(2x)^2 \geq 0$

$x\in(0,\sqrt{3})$

$$I = \int_0^{\sqrt{3}} \sqrt{1+\left(\frac{2\sqrt{12-4x^2}}{4}\right)^2}\, dx$$

$$= \int_0^{\sqrt{3}} \sqrt{4-x^2}\, dx$$

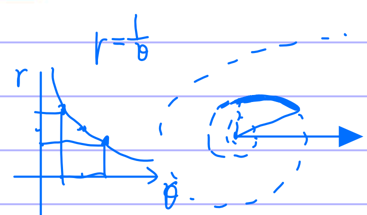
$$\overset{x=2\sin t}{=} \int_0^{\frac{\pi}{3}} 4\cos^2 t\, dx = 2\int_0^{\frac{\pi}{3}} 1+\cos 2t\, dt = 2\left[\frac{\pi}{3} + \frac{\sin 2t}{2}\Big|_0^{\frac{\pi}{3}}\right] = \frac{3\sqrt{3}+4}{6}\pi$$

$\frac{\sqrt{3}}{4}\pi \qquad \frac{\sqrt{3}}{8}\pi + \frac{\pi}{6} \qquad \sin\frac{2}{3}\pi$

例 10.20

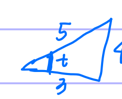
例 10.20 求曲线 $r\theta=1$ 自 $\theta=\frac{3}{4}$ 至 $\theta=\frac{4}{3}$ 一段的弧长.

$r = \frac{1}{\theta}$



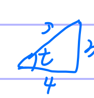
$$\int_{\frac{3}{4}}^{\frac{4}{3}} \sqrt{r^2 + (r'_\theta)^2}\, d\theta$$

$$= \int_{3/4}^{4/3} \sqrt{\frac{1}{\theta^2}+\frac{1}{\theta^4}}\, d\theta = \int_{3/4}^{4/3} \frac{\sqrt{1+\theta^2}}{\theta^2}\, d\theta$$



$$\overset{\theta=\tan t}{=} \int_{\arctan\frac{3}{4}}^{\arctan\frac{4}{3}} \frac{\sec^3 t}{\tan^2 t}\, d\theta = \ln|\sec t + \tan t| - \frac{1}{\sin t}\Big|_{\arctan\frac{3}{4}}^{\arctan\frac{4}{3}}$$

$$= \ln\frac{\frac{5}{3}+\frac{4}{3}}{\frac{5}{4}+\frac{3}{4}} - \left(\frac{5}{4}-\frac{5}{3}\right)$$

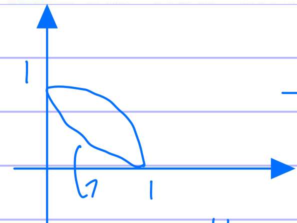


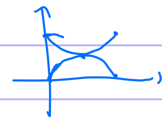
$$= \ln\frac{3}{2} + \frac{5}{12}$$

★★★

例10.23

例 10.23 设 D 是由曲线 $y=\sqrt{1-x^2}\,(0\leqslant x\leqslant 1)$ 与 $\begin{cases}x(t)=\cos^3 t,\\ y(t)=\sin^3 t\end{cases}\left(0\leqslant t\leqslant \frac{\pi}{2}\right)$ 围成的平面区域，求 D 绕 x 轴旋转一周所得旋转体的体积和表面积.





$(\cos^3 t)' = -3\cos^2 t\sin t < 0.$ $\quad\tan t=\sqrt{\frac{1}{2}}\quad t=\frac{\pi}{4}$ 时

$(-3\cos^2 t\sin t)' = +6\cos t\cdot\sin^2 t - 3\cos^3 t = \cos t(6\sin^2 t - 3\cos^2 t)$

① $V = \frac{2}{3}\pi\cdot 1^2 - \int_0^1 \pi y^2\,dx$

换元法 $\Longrightarrow$

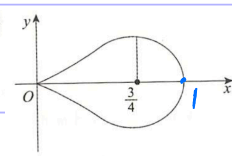
$= \frac{2}{3}\pi + \int_{\cancel{0}\,\frac{\pi}{2}}^{\cancel{\frac{\pi}{2}}\,0} \pi\sin^6 t\cdot 3\cos^2 t\cdot\sin t\,dt = \frac{2}{3}\pi + 3\pi\int_{\cancel{0}\,\frac{\pi}{2}}^{\cancel{\frac{\pi}{2}}\,0} \sin^9 t - \sin^7 t\;dt$

② 套公式 $\Longrightarrow$

$S = 2\pi + \int_0^{\frac{\pi}{2}} 2\pi\, y(t)\sqrt{(x_t')^2+(y_t')^2}\;dt$

例10.24

例 10.24 求由曲线 $y^2=x^3-x^4$ 所围成的平面图形的形心.



$y = x\cdot\sqrt{x-x^2} = x\sqrt{x(1-x)}$

$x-x^2>0 \quad x\in(0,1)$

$\bar{y}=0$

$\frac{6-8}{4-6}$ $\quad$ 10-2,1

$\bar{x} = \dfrac{\iint_D x\,d\sigma}{\iint_D 1\,d\sigma} = \dfrac{\int_0^1 dx\int_0^y x\,dy}{\int_0^1 dx\int_0^y 1\,dy} = \dfrac{\int_0^1 xy\,dx}{\int_0^1 y\,dx} = \dfrac{\int_0^1 x^2\sqrt{x(1-x)}\,dx}{\int_0^1 x\sqrt{x(1-x)}\,dx} \xlongequal{x=\sin^2 t} \dfrac{\int_0^{\frac{\pi}{2}} \sin^5 t\cdot\cos^2 t\cdot 2\sin t\,dt}{\int_0^{\frac{\pi}{2}} \sin^3 t\cdot\cos^2 t\cdot 2\sin t\,dt}$

$= \dfrac{\left(1-\frac{7}{8}\right)\frac{5}{6}\left(\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}\right)}{\left(1-\frac{5}{6}\right)\left(\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}\right)} = \frac{5}{8}$

形心 $\left(\frac{5}{8}, 0\right)$

例10.25

例 10.25 设函数 $y=f(x)$ 在区间$[0,1]$上非负、存在二阶导数，且 $f(0)=0$，有一块质量均匀分布的平板 D，其占据的区域是曲线 $y=f(x)$ 与直线 $x=1$ 以及 x 轴围成的平面图形. 用 $\bar{x}$ 表示平板 D 的质心的横坐标. 证明：

(1) 若 $f'(x)>0\,(0\leqslant x\leqslant 1)$，则 $\bar{x}>\frac{1}{2}$（见图 10-17 或见图 10-18）；

(2) 若 $f''(x)>0\,(0\leqslant x\leqslant 1)$，则 $\bar{x}>\frac{2}{3}$（见图 10-18）.

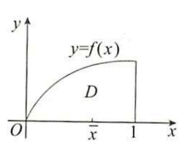


图 10-17

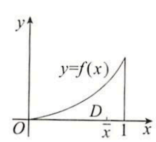


图 10-18

(1) 证 $\bar{x} = \dfrac{\int_0^1 x f(x)\,dx}{\int_0^1 f(x)\,dx} > \dfrac{1}{2}$

$f(x)$

$f(x)\int_0^x t f(t)\,dt = \int_0^x t\,dF(t) = \left[tF(t)\Big|_0^x - \int_0^x F(t)\,dt\right] f(x)$

$= x\int_0^x f(t)\,dt$

证 $= \dfrac{\int_0^x t f(t)\,dt}{\int_0^x f(t)\,dt} > \dfrac{x}{2}$

证 $g(x)\ x\int_0^x f(t)\,dt - 2\int_0^x t f(t)\,dt < 0$

$\dfrac{F(x)-F(0)}{x-0} = f(\xi)$

$g'(x) = F(x) + xf(x) - 2xf(x) = F(x) - xf(x) = x(f(\xi) - f(x)) < 0$

$g(x)\downarrow \quad g(0) = 0$ ※

(2) 证 $\dfrac{\int_0^1 x f(x)\,dx}{\int_0^1 f(x)\,dx} > \dfrac{2x}{3}$

$g(x) = 3\int_0^x t f(t)\,dt - 2x\int_0^x f(t)\,dt$

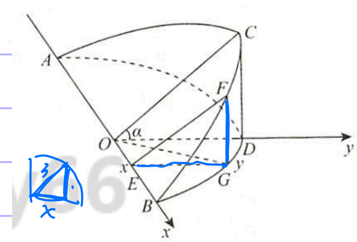
$g'(x) = 3xf(x) - 2F(x) - 2xf(x) = xf(x) - 2F(x)$

$g''(x) = xf'(x) - f(x) = x[f'(x) - f'(\xi)] > 0$

$g'(x)\uparrow \quad g'(0) = 0 \quad g'(x) > 0 \quad g(x)\uparrow \quad g(0) = 0 \quad g(x) > 0$ ※

例 10.26

例 10.26 设一个底面半径为 3 的圆柱体，被一个与圆柱的底面相交、夹角为$\frac{\pi}{4}$ 且过底面直径 AB 的平面所截，求截下的楔形体的体积.

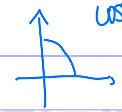


$V = \int s\,dx$

$= 2\int_0^3 s\,dx \Rightarrow V = \int_0^3 9 - x^2\,dx$

$S = \dfrac{(\sqrt{9-x^2})^2}{2} \qquad = 18$

Part Two 其它例题



例10.1

例 10.1 设 $P(x,y)$ 为曲线

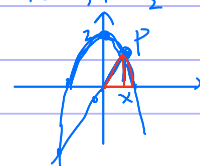
$$L:\begin{cases}x=\cos t,\\ y=2\sin^2 t\end{cases}\left(0\leqslant t\leqslant\frac{\pi}{2}\right)$$

上一点,作过原点 $O(0,0)$ 和点 P 的直线 OP,将曲线 L、直线 OP 以及 x 轴所围成的平面图形记为 A.

(1) 求平面图形 A 的面积 $S(x)$ 的表达式;

(2) 将平面图形 A 的面积 $S(x)$ 表示为 t 的函数 $S=S_1(t)$,并求 $\frac{dS_1}{dt}$ 取得最大值时点 P 的坐标.

(1) $x=\sqrt{1-\frac{y}{2}}$ $\quad y=-2x^2+2$



设 $P(t,-2t^2+2)$

$$\begin{cases}y=-2x^2+2\\ y=\dfrac{-2t^2+2}{t}x\end{cases}$$

$x^2-\frac{(t^2+1)}{t}x-1=0$

$x_{1,2}=\dfrac{\frac{t^2+1}{t}\pm\sqrt{(\frac{t^2+1}{t})^2+4}}{2}=\frac{1}{2}\left[(t+\frac{1}{t})\pm(t+\frac{1}{t})\right]$

$x_1=t \quad x_2=-\frac{1}{t}$

$$S(x)=\int_0^x\frac{-2x^2+2}{x}t\,dt+\int_x^1(-2t^2+2)\,dt$$
$$=\frac{-x^2+1}{x}t^2\Big|_0^x+\left(-\frac{2}{3}x^3+2t\right)\Big|_x^1$$
$$=-x^3+x+\left(-\frac{2}{3}+2\right)-\left(-\frac{2}{3}x^3+2x\right)$$
$$=-\frac{x^3}{3}+\frac{4}{3}-x$$

(2) $S(t)=-\frac{\cos^3 t}{3}+\frac{4}{3}-\cos t$

$S'(t)=+\cos^2 t\sin t+\sin t=2\sin t-\sin^3 t=\sin t(2-\sin^2 t)$ $\quad t=0$

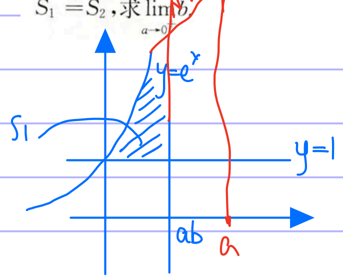
$S''(t)=\cos t(2-\sin^2 t)+\sin t\cdot(-2\sin t)\cdot\cos t$

$=2\cos t-3\cos t\sin^2 t=\cos t(2-3\sin^2 t)=0$ $\quad \sin t=\sqrt{\frac{2}{3}}$

$x=\frac{1}{\sqrt{3}}$ $\quad y=\frac{4}{3}$ $\quad P(\frac{1}{\sqrt{3}},\frac{4}{3})$

例10.2

例 10.2 设 $a>0,0<b<1$,由曲线 $y=e^x$,直线 $y=1$ 与直线 $x=ab$ 所围平面区域的面积记为 S_1,由曲线 $y=e^x$,直线 $x=ab$ 与直线 $y=e^a$ 所围平面区域的面积记为 S_2,若 $S_1=S_2$,求 $\lim\limits_{a\to0^+}b$.

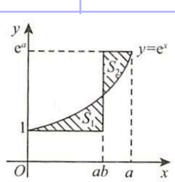


$S_1=\int_0^{ab}e^x-1\,dx=e^x-x\Big|_0^{ab}=e^{ab}-ab-1$

$\int_{ab}^{a}(e^a-e^{ax})\,dx$

$S_2=\cancel{\int_0^{ab}|e^x-e^{ax}|dx=\left|e^x-\frac{e^{ax}}{a}\right|\Big|_0^{ab}=\left|e^{ab}-\frac{e^{a^2b}}{a}\right|-\left|1-\frac{1}{a}\right|}$

$=(e^{ab}-1)\frac{|a-1|}{a}$

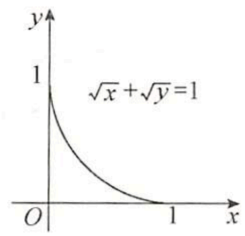


$(e^{ab}-1)-ab=(e^{ab}-1)\frac{|a-1|}{a}$

$\lim\limits_{a\to0^+}b=\lim\limits_{a\to0^+}\dfrac{(e^{ab}-1)(1-\frac{|a-1|}{a})}{a}=\lim\limits_{a\to0^+}b\cdot(1-\frac{1}{a})$ $\quad \cdots=\frac{1}{2}$

例10.4

例 10.4 求曲线$\sqrt{x}+\sqrt{y}=1$与坐标轴所围图形的面积.

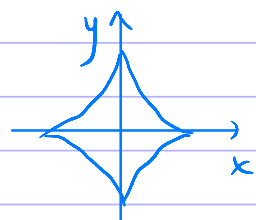


$y=(1-\sqrt{x})^2 = 1+x-2\sqrt{x}$

$\int_0^1 (1-\sqrt{x})^2\,dx = x+\frac{x^2}{2}-\frac{4}{3}x^{\frac{3}{2}}\Big|_0^1 = \frac{1}{6}$

例10.6

例 10.6 求星形线$\begin{cases}x=\cos^3 t,\\ y=\sin^3 t\end{cases}$所围图形的面积.



$S=4\int_0^1 y\,dx = 4\int_0^{\frac{\pi}{2}} \sin^4 t\cdot 3\cos^2 t\,dt$

$=12\int_0^{\frac{\pi}{2}} \sin^4 t - \sin^6 t\,dt = 12\cdot\left(1-\frac{5}{6}\right)\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{3}{8}\pi$

例10.8

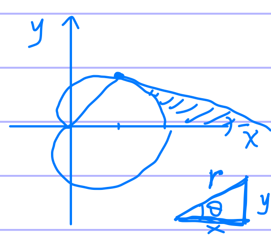
例 10.8 求阿基米德螺线$r=a\theta(a>0)$的第一圈与极轴所围图形的面积.

$S=\int_0^{2\pi}\frac{1}{2}a^2\theta^2\,d\theta = \frac{1}{6}a^2\theta^3\Big|_0^{2\pi} = \frac{a^2}{6}8\pi^3 = \frac{4}{3}a^2\pi^3$

例10.9

$\frac{2+\sqrt{2}}{2}$

例 10.9 曲线$r=1+\cos\theta$与其在点$\left(\frac{\pi}{4},1+\frac{\sqrt{2}}{2}\right)$处的切线及$x$轴所围图形面积为______.



$\frac{dy}{dx}=\frac{d\,r\sin\theta}{d\,r\cos\theta}=\frac{\sin\theta\cdot dr}{-r\sin\theta\,d\theta}=-\frac{1}{r}\frac{dr}{d\theta}$

$\frac{dr}{d\theta}=-\sin\theta \Rightarrow \frac{dy}{dx}=\frac{\sin\theta}{r}\Big|_{(\frac{\pi}{4},1+\frac{\sqrt{2}}{2})}=\frac{1}{1+\sqrt{2}}$

$r=1+\cos\theta$

$\begin{cases}y=r\sin\theta\\ x=r\cos\theta\end{cases}$ $\theta=\arctan\frac{y}{x}$ $r=\sqrt{x^2+y^2}$

$\sqrt{x^2+y^2}=1+\frac{x}{\sqrt{x^2+y^2}}$

$x^2+y^2=\sqrt{x^2+y^2}+x$

$x=y=\sqrt{\frac{(1+\frac{\sqrt{2}}{2})^2}{2}}=\frac{\frac{2+\sqrt{2}}{2}}{\sqrt{2}}=\frac{1+\sqrt{2}}{2}$

$2x+2y\cdot y'=\frac{1}{2}(x^2+y^2)^{-\frac{1}{2}}\cdot(2x+2y\cdot y')+1$

$(1+\sqrt{2})(1+y')=\frac{1}{\sqrt{2}(1+\sqrt{2})}\cdot(1+\sqrt{2})(1+y')+1$

$\frac{1+\frac{\sqrt{2}}{2}}{2}$

$1=\left(1-\frac{1}{\sqrt{2}(1+\sqrt{2})}\right)(1+y')(1+\sqrt{2})$

$\frac{\sqrt{2}}{1+\sqrt{2}}=1+y' \Rightarrow y'=\frac{-1}{1+\sqrt{2}}$

$\frac{y-y_0}{x-x_0}=\frac{-1}{1+\sqrt{2}}$ $y=\frac{-1}{1+\sqrt{2}}(x-x_0)+y_0 \Rightarrow y=0 \Rightarrow x=\frac{4+3\sqrt{2}}{2}$

$$S_{\text{底}} = \frac{1}{2}\,\frac{4+3\sqrt{2}}{2}\,\frac{1+\sqrt{2}}{2} = \frac{10+7\sqrt{2}}{8}$$

$$S = \frac{10+7\sqrt{2}}{8} - \int_0^{\frac{\pi}{4}} r^2 d\theta = \frac{1}{8} + \frac{\sqrt{2}}{2} + \frac{3}{16}\pi$$

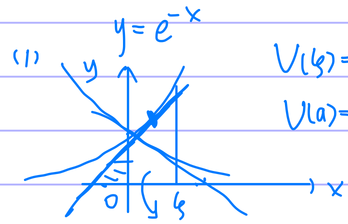
例 10.10

例 10.10 设曲线方程为 $y = e^{-x}(x \geqslant 0)$.

(1) 把曲线 $y = e^{-x}$, x 轴, y 轴和直线 $x = \xi(\xi > 0)$ 所围平面图形绕 x 轴旋转一周, 得一旋转体, 求此旋转体体积 $V(\xi)$, 并求出满足 $V(a) = \frac{1}{2}\lim\limits_{\xi \to +\infty} V(\xi)$ 的 a;

(2) 在此曲线上找一点, 使过该点的切线与两个坐标轴所围平面图形的面积最大, 并求出该面积.

(1)



$$V(\xi) = \int_0^{\xi} \pi e^{-2x} dx = \frac{\pi}{-2} e^{-2x}\Big|_0^{\xi} = -\frac{\pi}{2}(e^{-2\xi} - 1)$$

$$V(a) = \frac{1}{2} \cdot \lim_{\xi \to +\infty} V(\xi) = \frac{1}{2} \cdot \left(-\frac{\pi}{2}\right) \cdot (-1) = \frac{\pi}{4} = -\frac{\pi}{2}(e^{-2a} - 1) \Rightarrow a = \frac{\ln 2}{2}$$

(2)

$$\frac{y - e^{-x_0}}{x - x_0} = -e^{-x_0} \qquad y\big|_{x=0} = x_0 e^{-x_0} + e^{-x_0}$$

$$x\big|_{y=0} = 1 + x_0$$

$$S(x) = \frac{1}{2}(1+x)^2 e^{-x}$$

$$S'(x) = e^{-x}\left(1 + x - \frac{1}{2}(1+x)^2\right) = 0$$

$$\Rightarrow 2 + 2x = (1+x)^2$$

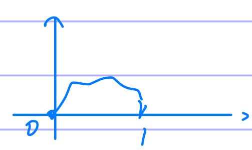
$$2 + 2x = 1 + x^2 + 2x \Rightarrow x^2 = 1 \Rightarrow x = \pm 1 \Rightarrow x = 1$$

$$S(1) = 2e^{-1}$$

例 10.12

例 10.12 求曲线 $y = \sqrt{x(1-x)^9}$ 在 $[0,1]$ 上与 x 轴所围图形绕 x 轴旋转一周所得的旋转体的体积.

$$y = \sqrt{x(1-x)^9}$$



$$V = \int_0^1 \pi x(1-x)^9 dx$$

$$= \int_0^1 \pi (1-x) x^9 dx = \pi \int_0^1 x^9 - x^{10} dx = \pi\left[\frac{x^{10}}{10} - \frac{x^{11}}{11}\right]_0^1 = \pi\left(\frac{1}{10} - \frac{1}{11}\right) = \pi \frac{1}{110}$$

例 10.13

例 10.13 已知函数 $f(x,y)$ 满足 $\frac{\partial f(x,y)}{\partial y} = 2(y+1)$, 且

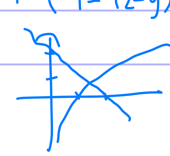
$$f(y,y) = (y+1)^2 - (2-y)\ln y,$$

求曲线 $f(x,y) = 0$ 所围图形绕直线 $y = -1$ 旋转一周所成旋转体的体积.

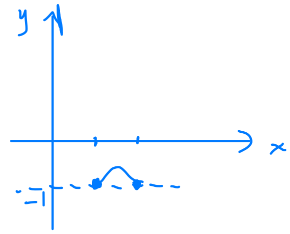
$$f(x,y) = y^2 + 2y + \varphi(x)$$

$$f(y,y) = y^2 + 2y + \varphi(y) = y^2 + 2y + (1 - (2-y)\ln y)$$

$$f(x,y) = (y+1)^2 - (2-x)\ln x = 0$$



$$y = \sqrt{(2-x)\ln x} - 1$$



$$y' = \frac{\frac{2}{x} - \ln x - 1}{2\sqrt{(2-x)\ln x}} = \frac{2 - x(\ln x + 1)}{2x\sqrt{(2-x)\ln x}} > 0 \quad x \in (1,2)$$

$2e-1-1$

$$V = \int_1^2 \pi (2-x)\ln x \, dx$$

$\int_1^2 \ln x \, d\frac{x^2}{2}$

$\ln x \quad \frac{1}{x}$

$x \quad \frac{x^2}{2}$

$$= \pi \left[2\int_1^2 \ln x \, dx - \int_1^2 x \ln x \, dx \right]$$

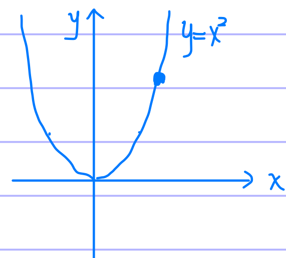
$$= \pi \left[2\left(x\ln x \Big|_1^2 - \int_1^2 1 \, dx \right) - \left(\frac{x^2}{2}\ln x \Big|_1^2 - \int_1^2 \frac{x}{2} \, dx \right) \right]$$

$$= \pi \left[2(2\ln 2 - 1) - \left(2\ln 2 - \frac{3}{4}\right) \right]$$

$$= \pi \left(2\ln 2 - \frac{5}{4} \right)$$

例 10.21

例 10.21 设 k 表示曲线 $y = x^2 (0 \leqslant x < +\infty)$ 在任一点 (x, y) 处的曲率，s 表示曲线在对应于区间 $[0, x]$ 上的一段弧长. 求导数 $\frac{dk}{ds}$.



$$y' = 2x \qquad k = \frac{|y''|}{\left(1 + (y')^{\frac{3}{2}}\right)^2} = \frac{2}{\left(1 + (2x)^{\frac{3}{2}}\right)^2}$$

$$y'' = 2$$

$$S = \int_0^x \sqrt{1 + 4t^2} \, dt \qquad = 2\left(1 + (2x)^{\frac{3}{2}}\right)^{-2}$$

$$\frac{dk}{ds} = \frac{dk}{dx} \frac{dx}{ds} = \frac{-4\left(1 + (2x)^{\frac{3}{2}}\right)^{-3} \cdot 3(2x)^{\frac{1}{2}}}{1 + 4x^2}$$

$$= -12 \frac{(2x)^{\frac{1}{2}}}{(1 + 4x^2)\left(1 + (2x)^{\frac{3}{2}}\right)^3}$$

例 10.22

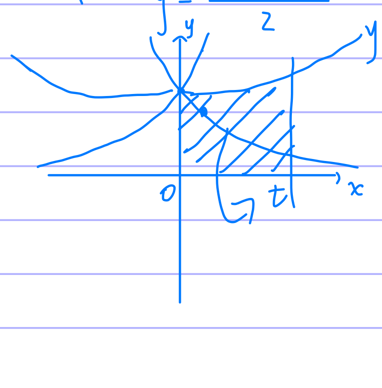
例 10.22 曲线 $y = \frac{e^x + e^{-x}}{2}$ 与直线 $x = 0, x = t (t > 0)$ 及 $y = 0$ 围成一曲边梯形. 该曲边梯形绕 x 轴旋转一周得一旋转体，其体积为 $V(t)$，侧面积为 $S(t)$，在 $x = t$ 处的底面积为 $F(t)$.

(1) 求 $\frac{S(t)}{V(t)}$ 的值；

(2) 计算极限 $\lim\limits_{t \to +\infty} \frac{S(t)}{F(t)}$.

$$y' = \frac{e^x - e^{-x}}{2}$$

(1) $y = \frac{e^x + e^{-x}}{2}$



$$V(t) = \int_0^t \pi \left(\frac{e^x + e^{-x}}{2} \right)^2 dx$$

$$S(t) = \int_0^t 2\pi \left(\frac{e^x + e^{-x}}{2} \right) \sqrt{1 + \left(\frac{e^x - e^{-x}}{2} \right)^2} \, dx$$

$$= \int_0^t 2\pi \left(\frac{e^x + e^{-x}}{2} \right)^2 dx$$

$$\frac{S(t)}{V(t)} = 2$$

$$\frac{e^{2x}-e^{-2x}+2x}{8} \qquad \frac{e^{2x}+e^{-2x}+2}{4}$$

$$\lim_{t\to+\infty}\frac{S(t)}{F(t)} = \lim_{t\to+\infty}\frac{\int_0^t 2\pi\left(\frac{e^x+e^{-x}}{2}\right)^2dx}{\pi\left(\frac{e^t+e^{-t}}{2}\right)^2} = \lim_{t\to+\infty}\frac{(e^{2t}-e^{-2t}+2t)-(e^2-e^{-2})}{e^{2t}+e^{-2t}+2} = 1$$

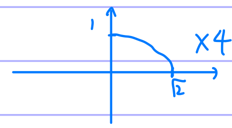
习10.1

10.1 （仅数学一、数学二）记 l_1 为椭圆 $x^2+2y^2=2$ 的周长，l_2 为曲线 $y_1=\sin x$ 在 $0\leqslant x\leqslant 2\pi$ 上的弧长，l_3 为曲线 $y_2=\frac{1}{2}\sin 2x$ 在 $0\leqslant x\leqslant 2\pi$ 上的弧长，则(　　).

(A) $l_1>l_2=l_3$　　(B) $l_1=l_2<l_3$

(C) $l_2>l_3=l_1$　　(D) $l_1=l_2=l_3$

$$\frac{x^2}{2}+y^2=1$$



$$y=\sqrt{1-\frac{x^2}{2}} \qquad y'=\frac{-x}{2\sqrt{1-\frac{x^2}{2}}}$$

$$l_1=4\int_0^{\sqrt{2}}\sqrt{1+\frac{x^2}{4(1-\frac{x^2}{2})}}\,dx = 4\int_0^{\sqrt{2}}\sqrt{1+\frac{x^2}{4-2x^2}}\,dx$$

$$l_2=\int_0^{2\pi}\sqrt{1+\cos^2x}\,dx = \int_0^{2\pi}\sqrt{1+\cos^2x}\,dx = 4\int_0^{\frac{\pi}{2}}\sqrt{1+\cos^2x}\,dx \xlongequal{x=\frac{\pi}{2}-x} 4\int_0^{\frac{\pi}{2}}\sqrt{1+\sin^2x}\,dx$$

$$l_3=\int_0^{2\pi}\sqrt{1+\cos^2 2x}\,dx \xlongequal{2x=t} \frac{1}{2}\int_0^{4\pi}\sqrt{1+\cos^2t}\,dt = \int_0^{2\pi}\sqrt{1+\cos^2t}\,dt$$

$$\begin{cases} x=\sqrt{2}\cos t \\ y=\sin t \end{cases} \quad t\in[0,2\pi] \qquad l_1=\int_0^{2\pi}\sqrt{2\sin^2t+\cos^2t}\,dt = \int_0^{2\pi}\sqrt{1+\sin^2t}\,dt$$

$$= 4\int_0^{\frac{\pi}{2}}\sqrt{1+\sin^2t}\,dt$$

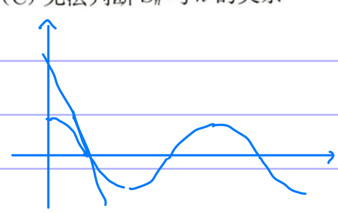
$\Rightarrow$ D

习 10.2

10.2 设 n 为正整数，S_n 是在第一象限内曲线 $y=n\cos nx$ 与该曲线在点 $\left(\frac{\pi}{2n},0\right)$ 处的切线所围成的平面图形的面积，则(B).

(A) S_n 与 n 有关　　(B) S_n 与 n 无关

(C) 无法判断 S_n 与 n 的关系　　(D) 以上结论都不正确



$$y\left(\frac{\pi}{2n}\right)=n\cdot\cos\frac{\pi n}{2n}=0$$

$$y'=-n^2\sin nx \qquad y'\left(\frac{\pi}{2n}\right)=-n^2$$

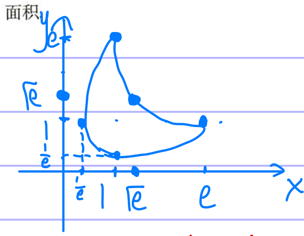
$$\frac{y-0}{x-\frac{\pi}{2n}}=-n^2 \qquad y=-n^2\left(x-\frac{\pi}{2n}\right) \xrightarrow{x=0} y=\frac{n\pi}{2}$$

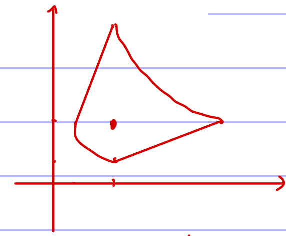
$$S=\frac{n\pi}{4}\cdot\frac{\pi}{2n}-\int_0^{\frac{\pi}{2n}}n\cos nx\,dx$$

$$=\frac{\pi^2}{8}-\sin nx\Big|_0^{\frac{\pi}{2n}}=\frac{\pi^2}{8}-1$$

习10.3

10.3 求由曲线 $|\ln x|+|\ln y|=1$ 所围成的平面图形的面积.





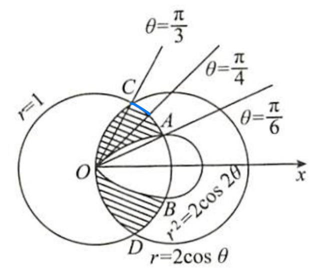
$$|\ln y| = 1-|\ln x| = \begin{cases} 1+\ln x, & \frac{1}{e}\le x<1 \\ 1-\ln x, & 1\le x\le e \end{cases}$$

$$y_1 = \begin{cases} ex, & \frac{1}{e}\le x<1 \\ \frac{e}{x}, & 1\le x\le e \end{cases} \qquad y_2 = \begin{cases} \frac{1}{ex}, & \frac{1}{e}\le x<1 \\ \frac{x}{e}, & 1\le x\le e \end{cases}$$

$$S = \int_{\frac{1}{e}}^{1} \left(ex-\frac{1}{ex}\right)dx + \int_{1}^{e}\left(\frac{e}{x}-\frac{x}{e}\right)dx$$

习10.4

10.4 求曲线 $r^2=2\cos 2\theta\left(-\frac{\pi}{4}\le\theta\le\frac{\pi}{4}\right)$, $r=2\cos\theta$, $r=1$ 所围成图形(见图10-20阴影部分)的面积.



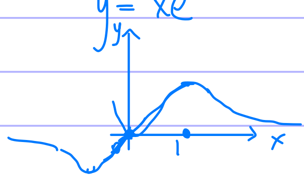
$$\cancel{S = 2\left[\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} 1-\sqrt{2\cos 2\theta}\,d\theta + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} 1\,d\theta + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 1-2\cos\theta\,d\theta\right]}$$

$$S = 2\left[\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} 1^2-2\cos 2\theta\,d\theta + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} 1^2\,d\theta + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 4\cos^2\theta\,d\theta\right]$$

习10.5

10.5 求曲线 $y=xe^{-\frac{x^2}{2}}$ 与其渐近线之间的面积.

$$y = xe^{-\frac{x^2}{2}} \qquad y' = e^{-\frac{x^2}{2}}(1-x^2) = e^{-\frac{x^2}{2}}\cdot(1-x)(1+x)$$



$$\lim_{x\to-\infty} e^{-\frac{x^2}{2}} = +\infty$$

$$\lim_{x\to+\infty} e^{-\frac{x^2}{2}} = 0 \quad \text{渐近线 } y=0$$

$$S = 2\int_0^{+\infty} xe^{-\frac{x^2}{2}}dx = 2\int_0^{+\infty} e^{-\frac{x^2}{2}}\,d\frac{x^2}{2} = 2\int_0^{+\infty} e^{-t}dt = -e^{-t}\Big|_0^{+\infty} = 2$$

习10.6

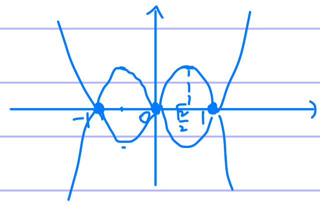
10.6 求由曲线 $y^2=x^2-x^4$ 所围成的平面图形的面积.

$y^2 = x^2(1-x^2)$

$y = \pm\left|x\sqrt{(1-x)(1+x)}\right|$

$y_1=\sqrt{x^2-x^4}$ $\quad x^2=\frac{1}{2}\quad x=\pm\frac{\sqrt{2}}{2}$

$y_1'=\dfrac{2x-4x^3}{2\sqrt{x^2-x^4}}=\dfrac{2x(1-2x^2)}{2\sqrt{x^2-x^4}}\overset{令}{=}0$



$$S=4\int_0^1\sqrt{x^2-x^4}\,dx$$
$$=4\int_0^1 x\sqrt{1-x^2}\,dx$$
$$=-2\int_0^1\sqrt{1-x^2}\,d(1-x^2)$$
$$=-2\cdot\frac{2}{3}(1-x^2)^{\frac{3}{2}}\Big|_0^1=\frac{4}{3}$$

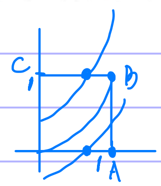
习10.7

10.7 设O为坐标原点，$A(1,0)$，$B(1,1)$，$C(0,1)$，记边长为1的正方形$OABC$内位于曲线 $y=x^2+t$（t为实数）下方图形的面积为$S(t)$.

(1) 求$S(t)$的表达式；

(2) $S(t)$在$[-1,1]$上是否满足拉格朗日中值定理的条件，说明理由.

(1)



$\begin{cases} y=x^2+t \\ y=1 \end{cases} \Rightarrow x=\sqrt{1-t}$

$\begin{cases} y=x^2+t \\ y=0 \end{cases} \Rightarrow x=\sqrt{-t}$

$$S(t)=\begin{cases} 0, & t\in(-\infty,-1) \\ \int_{\sqrt{-t}}^1 (x^2+t)\,dx, & t\in[-1,0) \quad =\frac{1}{3}+t+\frac{2}{3}(-t)^{\frac{3}{2}} \\ 1-\int_0^{\sqrt{1-t}} 1-(x^2+t)\,dx, & t\in[0,1) \quad =1-\frac{2}{3}(1-t)^{\frac{3}{2}} \\ 1, & t\in[1,+\infty) \end{cases}$$

(2) $\lim\limits_{t\to0^+}S'(t) = 1+(-t)^{\frac{1}{2}}(-1)\Big|_{t=0}=1$

$\lim\limits_{t\to0^-}S'(t) = -(1-t)^{\frac{1}{2}}(-1)\Big|_{t=0}=1$

满足

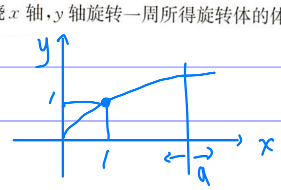
习10.8

10.8 求曲线 $y=\sin^4 x$ 在$[0,\pi]$上与x轴所围图形绕y轴旋转一周所得的旋转体的体积.

$$V=\int_0^\pi 2\pi x\sin^4 x\,dx=\int_0^\pi 2\pi(\pi-x)\sin^4 x\,dx$$
$$\Rightarrow V=\int_0^\pi \pi^2\sin^4 x\,dx=\pi^2\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}\times2=\frac{3}{8}\pi^3$$

习10.9

10.9 设D是由曲线 $y=x^{\frac{1}{3}}$，直线 $x=a(a>0)$ 及x轴所围成的平面图形，V_x，V_y分别是D绕x轴，y轴旋转一周所得旋转体的体积. 若$V_y=10V_x$，求a的值.



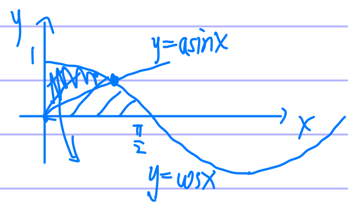
$$\int_0^a 2\pi x^{\frac{4}{3}}\,dx=10\int_0^a \pi x^{\frac{2}{3}}\,dx$$
$$2\pi\cdot\frac{3}{7}x^{\frac{7}{3}}\Big|_0^a=\frac{6\pi}{7}\cdot a^{\frac{7}{3}}=10\cdot\pi\cdot\frac{3}{5}x^{\frac{5}{3}}\Big|_0^a=6\pi\cdot a^{\frac{5}{3}}$$
$$a^{\frac{2}{3}}=7 \qquad a=7^{\frac{3}{2}}$$

习10.10

10.10 设由曲线 $y=\cos x\left(0\leqslant x\leqslant\frac{\pi}{2}\right)$ 及 x 轴、y 轴所围成平面图形的面积被曲线 $y=a\sin x(a>0)$ 二等分.

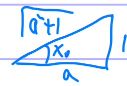
(1) 求 a 的值;

(2) 求曲线 $y=\cos x, y=a\sin x, x=0$ 所围平面图形绕 $y=0$ 旋转一周所成旋转体的体积.



$a\sin x_0=\cos x_0$

$x_0=\arctan\frac{1}{a}$



$\frac{1}{2}\int_0^{\frac{\pi}{2}}\cos x\,dx=\int_0^{x_0}\cos x-a\sin x\,dx$

$\frac{\sin x}{2}\Big|_0^{\frac{\pi}{2}}=\sin x+a\cos x\Big|_0^{x_0}$

$\frac{1}{2}=\sin x_0+a\cos x_0-a$

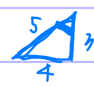
$\frac{1}{2}-\sin x_0=a(\cos x_0-1)$

$a=\frac{1-2\sin x_0}{2\cos x_0-2}=\frac{1-2\frac{1}{\sqrt{a^2+1}}}{2\frac{a}{\sqrt{a^2+1}}-2}=\frac{\sqrt{a^2+1}-2}{2a-2\sqrt{a^2+1}}=\frac{(\sqrt{a^2+1}-2)(a+\sqrt{a^2+1})}{2(a^2-a^2-1)}$

$\Rightarrow\ -2a=(a-2)\sqrt{a^2+1}-2a+a^2+1$

$2-a=\sqrt{a^2+1}$

$4+a^2-4a=a^2+1\qquad a=\frac{3}{4}$



(2) $V=\int_0^{\arctan\frac{4}{3}}\pi\left(\cos^2x-\frac{9}{16}\sin^2x\right)dx$ $\qquad\frac{12}{25}$

$2\sin x\cos x$

$=\pi\int_0^{\arctan\frac{4}{3}}\frac{1+\cos 2x}{2}-\frac{9}{16}\frac{1-\cos 2x}{2}\,dx$

$=\pi\left[\frac{x}{2}+\frac{\sin 2x}{4}-\frac{9}{16}\left(\frac{x}{2}-\frac{\sin 2x}{4}\right)\right]_0^{\arctan\frac{4}{3}}$

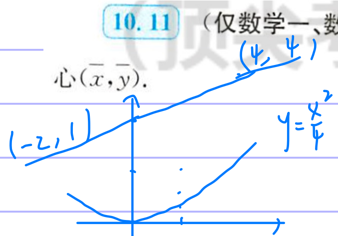
$=\pi\left[\frac{1}{2}\arctan\frac{4}{3}+\frac{1}{4}\frac{24}{25}-\frac{9}{16}\left(\frac{1}{2}\arctan\frac{4}{3}-\frac{1}{4}\frac{24}{25}\right)\right]$

$=\pi\left[\frac{7}{32}\arctan\frac{4}{3}+\frac{7}{16}\frac{1}{4}\frac{24}{25}\right]$

$=\frac{\pi}{64}\left[14\arctan\frac{4}{3}+7\frac{24}{25}\right.$

习10.11

10.11 (仅数学一、数学二)求抛物线 $y=\frac{x^2}{4}$ 与直线 $y=\frac{1}{2}x+2$ 所围平面图形 D 的形心 $(\bar{x},\bar{y})$.



$\bar{x}=\frac{\iint x\,d\sigma}{\iint d\sigma}=\frac{\int_{-2}^{4}x\left(\frac{x}{2}+2-\frac{x^2}{4}\right)dx}{\int_{-2}^{4}\frac{x}{2}+2-\frac{x^2}{4}\,dx}=\frac{\frac{x^3}{6}+x^2-\frac{x^4}{16}\Big|_{-2}^{4}}{\frac{x^2}{4}+2x-\frac{x^3}{12}\Big|_{-2}^{4}}=\frac{9}{9}=1$

$\bar{y}=\frac{\iint y\,d\sigma}{\iint d\sigma}=\frac{\int_{-2}^{4}dx\int_{y_1}^{y_2}y\,dy}{9}=\frac{\int_{-2}^{4}\frac{1}{2}\left(\left(\frac{x}{2}+2\right)^2-\frac{x^4}{16}\right)dx}{9}=\frac{\frac{x^3}{12}+4x+x^2-\frac{x^5}{80}\Big|_{-2}^{4}}{9}$

$\frac{x^2}{4}=\frac{x}{2}+2$

$x^2-2x-8=0$

$x\quad 2\quad x_1=-2$

$x\quad -4\quad x_2=4$

$\frac{1}{2}\left[\frac{x^2}{4}+4+2x-\frac{x^4}{16}\right)$

$\frac{144}{90}=\frac{16}{10}=\frac{8}{5}$

$\frac{1}{32}\left[4x^2+64+32x-x^4\right)$

$672-211.2$

$\frac{1}{32}\left[\frac{4}{3}x^3+64x+16x^2-\frac{x^5}{5}\right]_{-2}^{4}=\frac{1}{32}\left[\frac{4}{3}(64+8)+64\times6+16\times12-\frac{1}{5}(1024+32)\right]=14.4$

$36\quad 72\quad 96$

1000.10.1

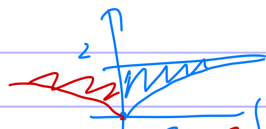
1. 曲线 $y=\int_0^x e^{-\sqrt{t}}dt$ 与 y 轴及其 $x\to+\infty$ 方向的水平渐近线所围图形的面积为（　　）.

(A)4　　(B)8　　(C)12　　(D)16

$-2e^{-u}$

$\lim\limits_{x\to+\infty}\int_0^x e^{-\sqrt{t}}dt=\lim\limits_{x\to+\infty}\int_0^{\sqrt{x}}2ue^{-u}du=\lim\limits_{x\to+\infty}\int_0^{\sqrt{x}}-2u\,de^{-u}=\lim\limits_{x\to+\infty}\left(-2ue^{-u}\Big|_0^{\sqrt{x}}+2\int_0^{\sqrt{x}}e^{-u}du\right)$

$=\lim\limits_{x\to+\infty}\underbrace{-2\sqrt{x}e^{-\sqrt{x}}-2e^{-\sqrt{x}}}+2=2$



$S=2\int_0^{+\infty}\sqrt{x}e^{-\sqrt{x}}+e^{-\sqrt{x}}dx=4\int_0^{+\infty}\underbrace{(t+t^2)e^{-t}dt}_{3}=12$

$t+t^2$　$1+2t$　2　0

e^{-t}　$-e^{-t}$　e^{-t}　$-e^{-t}$

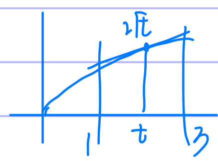
$e^{-t}(-t-t^2+1+t+2)$

1000.10.2

2. 设曲线 $y=2\sqrt{x}$ 与其上一点 $(t,2\sqrt{t})$ 处的切线以及直线 $x=1,x=3$ 围成的平面区域的面积记为 $A(t)$，这里 $t>0$，则当 $A(t)$ 取得最小值时相应切线的方程为（　　）.

(A) $y=\sqrt{2}x+\frac{1}{\sqrt{2}}$　　(B) $y=x+\frac{1}{2}$

(C) $y=\frac{x}{2}+2$　　(D) $y=\frac{x}{\sqrt{2}}+\sqrt{2}$



$\frac{y-2\sqrt{t}}{x-t}=\frac{1}{\sqrt{t}}$　　$y=\frac{1}{\sqrt{t}}(x-t)+2\sqrt{t}$

$S=\int_1^3\frac{x-t}{\sqrt{t}}+2\sqrt{t}-2\sqrt{x}\,dx=\frac{x^2}{2\sqrt{t}}+\sqrt{t}x-\frac{4}{3}x^{\frac{3}{2}}\Big|_1^3$

$=\left(\frac{9}{2\sqrt{t}}+3\sqrt{t}-4\sqrt{3}\right)-\left(\frac{1}{2\sqrt{t}}+\sqrt{t}-\frac{4}{3}\right)$

$=\frac{4}{\sqrt{t}}+2\sqrt{t}-4\sqrt{3}+\frac{4}{3}$

$S'=\frac{t-2}{\sqrt{t^3}}=0\Rightarrow t=2\Rightarrow D$

1000.10.1.3

3. 如图 1-10-1 所示，设立体的底是介于 $y=x^2-1$ 和 $y=0$ 之间的平面区域，它垂直于 x 轴的任一截面是一个等边三角形，则此立体的体积为（　　）.

(A) $\frac{\sqrt{3}}{15}$　　(B) $\frac{2\sqrt{3}}{15}$

(C) $\frac{3\sqrt{3}}{15}$　　(D) $\frac{4\sqrt{3}}{15}$

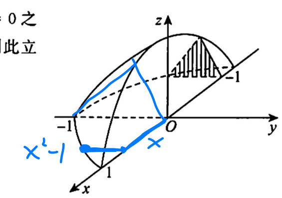
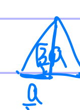
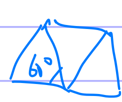


图 1-10-1



$\frac{\frac{\sqrt{3}}{2}a\cdot a}{2}$



$V=2\int_0^1\frac{\sqrt{3}}{4}(x^2-1)^2dx=\int_0^1\frac{\sqrt{3}}{2}(x^4+1-2x^2)dx=\frac{\sqrt{3}}{2}\left(\frac{x^5}{5}+x-\frac{2x^3}{3}\right)\Big|_0^1$

$=\frac{\sqrt{3}}{2}\left(\frac{1}{5}+1-\frac{2}{3}\right)=\frac{4\sqrt{3}}{15}$

$\frac{6}{5}-\frac{2}{3}=\frac{8}{15}$

1000.10.1.4

$\sqrt{x}\sqrt{2-x}$

4. 由曲线 $y=\sqrt{2x-x^2}$ 与直线 $y=x$ 围成的平面图形绕直线 $x=2$ 旋转一周得到的旋转体的体积为（　　）.

(A) $\frac{\pi^2}{2}+\frac{2\pi}{3}$　　(B) $\frac{\pi^2}{2}+\frac{4\pi}{3}$　　(C) $\frac{\pi^2}{2}-\frac{2\pi}{3}$　　(D) $\frac{\pi^2}{2}-\frac{4\pi}{3}$

$y_1=\sqrt{2(x+2)-(x+2)^2}$

$y_2=x+2$

$x=0$

$\Rightarrow$　$y_1=\sqrt{1-x^2}$

$y_2=1-x$

$V=\int_0^1 2\pi(x+1)\left(\sqrt{1-x^2}-(1-x)\right)dx$

$=2\pi\left[\int_0^1 x\sqrt{1-x^2}dx+\int_0^1\sqrt{1-x^2}dx-(x+1)(1-x)dx\right]$

$x^2=2x-x^2$
$2x=2 \quad x=1$

$$= -\pi\int_0^1 \sqrt{1-x^2}\, d(1-x^2) + 2\pi\int_0^1 \sqrt{1-x^2}\, dx - 2\pi\int_0^1 1-x^2\, dx$$
$$= -\pi (1-x^2)^{\frac{3}{2}}\cdot\frac{2}{3}\Big|_0^1 + 2\pi\cdot\frac{\pi}{4} - 2\pi\left(x-\frac{x^3}{3}\right)\Big|_0^1$$
$$= \frac{2\pi}{3} + \frac{\pi^2}{2} - 2\pi\cdot\frac{2}{3} = -\frac{2}{3}\pi + \frac{\pi^2}{2}$$

1000.10.1.5

5. 设曲线弧方程为 $y=\int_{-\frac{\pi}{2}}^{x}\sqrt{\cos t}\,dt\left(-\frac{\pi}{2}\leqslant x\leqslant\frac{\pi}{2}\right)$，则其弧长为______.

$y'=\sqrt{\cos x}$

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sqrt{1+\cos x}\,dx$

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sqrt{2}\cos\frac{x}{2}\,dx$

$2\sqrt{2}\times\sin\frac{x}{2}\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 2\sqrt{2}\cdot\frac{\sqrt{2}}{2}\times 2 = 4$

1000.10.1.6

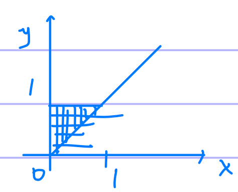
6. 曲线 $r=1+\cos\theta$ 介于 $0\leqslant\theta\leqslant\pi$ 的弧长为______.

$\int_0^{\pi}\sqrt{(1+\cos\theta)^2+\sin^2\theta}\,d\theta$

$=\int_0^{\pi}\sqrt{2+2\cos\theta} = \int_0^{\pi}2\cos\frac{\theta}{2}\,d\theta = 4\sin\frac{\theta}{2}\Big|_0^{\pi} = 4$

1000.10.1.7

7. $f(x)=\int_x^1\cos t^2\,dt$ 在区间$[0,1]$上的平均值为______.

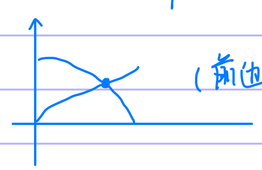


$\dfrac{\int_0^1 f(x)\,dx}{1-0} = \int_0^1 dy\int_0^y \cos y^2\,dx \qquad \sin y^2$

$= \int_0^1 y\cos y^2\,dy = \frac{1}{2}\int_0^1\cos y^2\,dy^2 = \frac{\sin y^2}{2}\Big|_0^1 = \frac{\sin 1}{2}$

1000.10.1.8

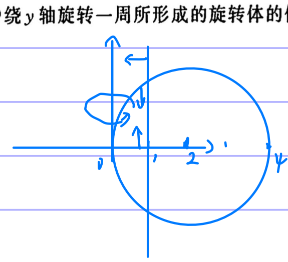
8. 设曲线 $y=\cos x\left(0\leqslant x\leqslant\frac{\pi}{2}\right)$ 与 x 轴，y 轴所围图形被曲线 $y=a\sin x(a>0)$ 分成面积相等的两部分，则常数 a 的值为 $\frac{3}{4}$.



（前面有）

1000.10.1.9

9. 设 $D=\{(x,y)\,|\,0\leqslant y\leqslant\sqrt{4x-x^2},x\leqslant 1\}$，则 D 绕 y 轴旋转一周所形成的旋转体的体积为



$y^2=4x-x^2=-(x-2)^2+4$

$y^2+(x-2)^2=4$

$V=\int_0^1 2\pi x\cdot\sqrt{4x-x^2}\,dx$

$=\int_0^1 2\pi x\sqrt{4-(x-2)^2}\,dx$

$\overset{x-2=t}{=}\int_{-2}^{-1}2\pi(t+2)\sqrt{4-t^2}\,dt$

$\overset{x=t+2,\ t=2\sin u}{=}\int_{-2}^{-1}\pi\sqrt{4-t^2}\,dt^2 + \int_{-2}^{-1}8\pi\cos u\,dt$

$-1 \quad = -\int_{-2}^{-1}\pi\sqrt{4-t^2}\,d(4-t^2) + \int_{-\frac{\pi}{2}}^{-\frac{\pi}{6}}16\pi\cos^2 u\,du$

$= -\pi\frac{2}{3}(4-t^2)^{\frac{3}{2}}\Big|_{-2}^{-1} + \int_{-\frac{\pi}{2}}^{-\frac{\pi}{6}}8\pi(1+\cos 2u)\,du$

$= -\pi\cdot 2\sqrt{3} + \frac{8}{3}\pi^2 + 8\pi\,\frac{\sin 2u}{2}\Big|_{-\frac{\pi}{2}}^{-\frac{\pi}{6}} = \frac{8}{3}\pi^2 - 2\sqrt{3}\pi + 4\pi\cdot\left(-\frac{\sqrt{3}}{2}\right)$

$= \frac{8}{3}\pi^2 - 4\sqrt{3}\pi$

1000.10.1.10 $c=0$

10. 设抛物线 $y = ax^2 + bx + c$ 通过 $(0,0)$ 和 $(1,2)$ 两点，其中 $a < -2$. 求 a,b,c 的值，使得该抛物线与曲线 $y = -x^2 + 2x$ 所围成区域的面积最小.

$2 = a+b \qquad b = 2-a$

$\Rightarrow \quad y = ax^2 + (2-a)x, \quad a<-2 \qquad \dfrac{-b}{2a} = \dfrac{a-2}{2a} = \dfrac{1}{2} - \dfrac{1}{a} \in \left(\dfrac{1}{2}, 1\right)$

$y = x(2-x)$

$\begin{cases} y = -x^2+2x \\ y = ax^2 + (2-a)x \end{cases} \qquad (a+1)x^2 - ax = 0$

$x[(a+1)x - a] = 0$

$\Rightarrow x_0 = \dfrac{a}{a+1}$

$$S = \int_0^{\frac{a}{a+1}} [ax^2+(2-a)x] - [2x-x^2] \, dx$$

$$= \int_0^{\frac{a}{a+1}} (a+1)x^2 - ax \, dx$$

$$= \frac{a+1}{3}x^3 - \frac{a}{2}x^2 \Big|_0^{\frac{a}{a+1}}$$

$$= \frac{a+1}{3}\frac{a^3}{(a+1)^3} - \frac{a}{2}\frac{a^2}{(a+1)^2}$$

$$= \frac{1}{(a+1)^2}\left[\frac{a^3}{3} - \frac{a^3}{2}\right] = -\frac{a^3}{6(a+1)^2} \qquad 2a-3a-3$$

a	$(-\infty,-3)$	-3	$(-3,-2)$
$f'(a)$	$-$	0	$+$
$f(a)$	↘		↗

$$S' = \frac{-3a^2 \cdot 6(a+1)^2 + a^3 \cdot 12(a+1)}{36(a+1)^4} = \frac{6(a+1)a^2(-3(a+1)+2a)}{36(a+1)^4} = \frac{-(a+1)a^2(3+a)}{6(a+1)^4}$$

$\Rightarrow a = -3$ ✓

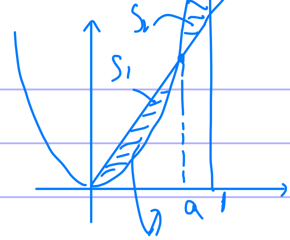
$b = 5$

1000.10.1.11

11. 设直线 $y = ax\,(0 < a < 1)$ 与抛物线 $y = x^2$ 所围成图形为 S_1，它们相交后的部分与直线 $x = 1$ 所围成图形为 S_2. 确定 a 的值，使平面图形 S_1 与 S_2 绕 x 轴旋转一周所得旋转体体积之和最小.

$y = ax$

$y = x^2$



$$V = \int_0^a \pi(a^2x^2 - x^4)\,dx + \int_a^1 \pi(x^4 - a^2x^2)\,dx$$

$$= \pi\left[\frac{a^2}{3}x^3 - \frac{x^5}{5}\right]_0^a + \pi\left[\frac{x^5}{5} - \frac{a^2}{3}x^3\right]_a^1$$

$$= \pi\left[\frac{a^5}{3} - \frac{a^5}{5} + \frac{1}{5} - \frac{a^2}{3} - \frac{a^5}{5} + \frac{a^5}{3}\right]$$

$$= \pi\left[\frac{4}{15}a^5 - \frac{a^2}{3} + \frac{1}{5}\right]$$

$$= 15\pi(4a^5 - 5a^2 + 3)$$

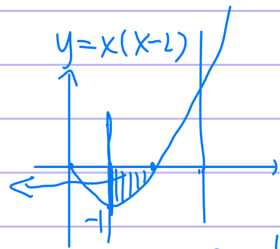
$$V' = 15\pi(20a^4 - 10a) = 150\pi(2a^4 - a) = 150\pi \cdot a \cdot (2a^3 - 1)$$

$a^3 = \frac{1}{2} \quad a = \sqrt[3]{\frac{1}{2}}$

1000. 10.1.12

12. 求曲线 $y=x^2-2x(1\leqslant x\leqslant 3)$，$y=0$，$x=1$ 与 $x=3$ 所围封闭图形面积，并求该平面图形绕 y 轴旋转一周所得旋转体体积.

$\frac{2}{3}-\frac{1}{4}$



$$V=2\pi\int_1^2 \cancel{x}(2x-x^2)dx+2\pi\int_2^3 \cancel{x}(x^2-2x)dx = 2\pi\left[\frac{2x^3}{3}-\frac{x^4}{4}\Big|_1^2+\frac{x^4}{4}-\frac{2x^3}{3}\Big|_2^3\right]$$

$$=2\pi\left[x^2-\frac{x^3}{3}\Big|_1^2+\frac{x^3}{3}-x^2\Big|_2^3\right] \qquad =2\pi\left[\left(\frac{16}{3}-4\right)2-\frac{5}{12}+\frac{81}{4}-18\right]$$

$$=2\pi\left[4-\frac{8}{3}-\frac{2}{3}+4-\frac{8}{3}\right]=4\pi \qquad =2\pi\left[\frac{32}{3}-26-\frac{5}{12}+\frac{81}{4}\right]=9\pi$$

$$S=\int_1^2(2x-x^2)\,dx+\int_2^3(x^2-2x)dx$$

$$=x^2-\frac{x^3}{3}\Big|_1^2+\frac{x^3}{3}-x^2\Big|_2^3$$

$$=2\left(4-\frac{8}{3}\right)-\frac{2}{3}=2$$

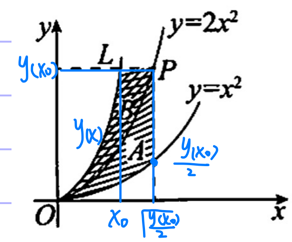
$x=y.$

$\sqrt{\frac{y(x_0)}{2}}=x^2$

1000. 10.1.13

13. (1) 如图 1-10-2 所示，设曲线 L 具有如下性质：中间曲线 $y=2x^2$ 上每一点 P 都使得图中 A 的面积等于 B 的面积. 求曲线 L 的方程；

(2) 如图 1-10-2 所示，A，B 绕 y 轴旋转一周所得的旋转体体积相等，求曲线 L 的方程.



(1)

$$S_A=\int_0^{\sqrt{\frac{y(x_0)}{2}}}x^2dx=\frac{x^3}{3}\Big|_0^{\sqrt{\frac{y(x_0)}{2}}}=\frac{1}{3}\left(\frac{y(x_0)}{2}\right)^{\frac{3}{2}}$$

$$S_B=\int_0^{x_0}y(x)-2x^2\,dx+\int_{x_0}^{\sqrt{\frac{y(x_0)}{2}}}y(x_0)-2x^2dx$$

$$=\int_0^{x_0}y(x)dx+\left(\sqrt{\frac{y(x_0)}{2}}-x_0\right)y(x_0)-2\int_0^{\sqrt{\frac{y(x_0)}{2}}}x^2dx$$

$$=\int_0^{x_0}y(x)dx+\left(\sqrt{\frac{y(x_0)}{2}}-x_0\right)y(x_0)-\frac{2}{3}\left(\frac{y(x_0)}{2}\right)^{\frac{3}{2}}$$

$$S_A=S_B\Rightarrow\int_0^{x_0}y(x)dx+\left(\sqrt{\frac{y(x_0)}{2}}-x_0\right)y(x_0)=\left(\frac{y(x_0)}{2}\right)^{\frac{3}{2}}$$

$$\int_0^x y\,dx+\left(\sqrt{\frac{y}{2}}-x\right)y=\left(\frac{y}{2}\right)^{\frac{3}{2}}$$

$$\int_0^x y\,dx+\frac{1}{2\sqrt{2}}y^{\frac{3}{2}}-xy=0$$

$$y+\frac{3}{4\sqrt{2}}y^{\frac{1}{2}}\cdot y'-y-xy'=0$$

$$\left(\frac{3\sqrt{2}}{8}y^{\frac{1}{2}}-x\right)y'=0\Rightarrow x=\frac{3\sqrt{2}}{8}y^{\frac{1}{2}}$$

(2)

(2) $V_A=\int_0^{\sqrt{\frac{y(x_0)}{2}}} 2\pi t^3\,dt = \frac{\pi t^4}{2}\Big|_0^{\sqrt{\frac{y(x_0)}{2}}} = \frac{\pi}{2}\cdot\frac{y^2(x_0)}{4} = \frac{\pi}{8}y^2(x_0)$

$V_B=\int_0^{x_0} 2\pi x\cdot(y(x)-2x^2)\,dx + \int_{x_0}^{\sqrt{\frac{y(x_0)}{2}}} 2\pi x\,(y(x_0)-2x^2)\,dx$

$= -2\int_0^{\sqrt{\frac{y(x_0)}{2}}} 2\pi x^3\,dx + \int_0^{x_0} 2\pi x\,y(x)\,dx + \int_{x_0}^{\sqrt{\frac{y(x_0)}{2}}} 2\pi x\,y(x_0)\,dx$

$= -\frac{\pi}{4}y^2(x_0) + \int_0^{x_0} 2\pi x\,dY(x) + \pi y(x_0)\cdot x^2\Big|_{x_0}^{\sqrt{\frac{y(x_0)}{2}}}$

$= \frac{\pi}{4}y^2(x_0) - \pi x_0^2 y(x_0) + 2\pi x Y(x)\Big|_0^{x_0} - \int_0^{x_0} 2\pi Y(x)\,dx$

$= \frac{\pi}{4}y^2(x_0) - \pi x_0^2 y(x_0) + 2\pi x_0 Y(x_0) - 2\pi\int_0^{x_0} Y(x)\,dx = \frac{\pi}{8}y^2(x_0)$

$\Rightarrow \frac{1}{8}y^2(x_0) - x_0^2 y(x_0) + 2x_0 Y(x_0) - 2\int_0^{x_0} Y(x)\,dx = 0$

$\frac{1}{4}y\cdot y' \underline{-2xy} - x^2 y' \underline{+2Y} \underline{+2xy} \underline{-2Y} = 0$

$(\frac{y}{4}-x^2)\,y' = 0$

$\Rightarrow y = 4x^2$

1000. 10.1.14

14. 如图 1-10-3 所示，阴影部分由曲线 $y=\sin x(0\leqslant x\leqslant\pi)$，直线 $y=a$ $(0\leqslant a\leqslant 1)$，$x=\pi$ 以及 y 轴围成. 此图形绕直线 $y=a$ 旋转一周形成旋转体. 问 a 为何值时，旋转体有最小体积、最大体积.

$\sin x = a$

$x = \arcsin a$

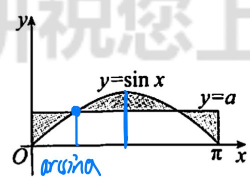


图 1-10-3

计算 $(0,\frac{\pi}{2})$ 体积 V

$V=\int_0^{\frac{\pi}{2}} \pi(a-\sin x)^2\,dx$

$=\pi\int_0^{\frac{\pi}{2}} a^2 - 2a\sin x + \frac{1-\cos 2x}{2}\,dx$

$=\pi\left[a^2x + 2a\cos x + \frac{x+\frac{\sin 2x}{2}}{2}\right]_0^{\frac{\pi}{2}}$

$=\pi\left[\frac{\pi}{2}a^2 - 2a + \frac{\frac{\pi}{2}}{2}\right] = \pi\left[\frac{\pi}{2}a^2 - 2a + \frac{\pi}{4}\right]$

$V' = \pi[\pi a - 2] \overset{\text{令}}{=} 0 \Rightarrow a = \frac{2}{\pi}$

a	0	$(0,\frac{2}{\pi})$	$\frac{2}{\pi}$	$(\frac{2}{\pi},1)$	1
V'		$-$	0	$+$	
V	$\frac{\pi^2}{4}$	$\searrow$	$\frac{\pi^2}{4}-2$	$\nearrow$	$\frac{3}{4}\pi^2-2\pi$

π

max $\frac{\pi^2}{4}$ $a=0$

min $\frac{\pi^2}{4}-2$ $a=\frac{2}{\pi}$

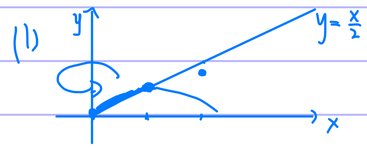
1000.10.1.15

15. 设由曲线 $y = y(x) = \lim\limits_{a \to -\infty} \dfrac{x}{1 + x^2 - e^{ax}}$ 和直线 $y = \dfrac{1}{2}x$ 围成的平面图形为 D，求：

(1) D 的面积 S；

(2) D 绕 y 轴旋转一周围成的旋转体的体积 V.

(1)



$$\lim_{a \to -\infty} \frac{x}{1+x^2-e^{ax}} = \frac{x}{1+x^2}$$

$$y' = \frac{1+x^2-2x^2}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$$

$$S = \int_0^1 \frac{x}{1+x^2} - \frac{x}{2}\,dx = \frac{1}{2}\int_0^1 \frac{d\,1+x^2}{1+x^2} - \int_0^1 \frac{x}{2}\,dx$$

$$= \frac{1}{2}\ln(1+x^2)\Big|_0^1 - \frac{x^2}{4}\Big|_0^1 = \frac{\ln 2}{2} - \frac{1}{4}$$

(2) $$V = \int_0^1 2\pi x\left(\frac{x}{1+x^2} - \frac{x}{2}\right)dx$$

$$= 2\pi\int_0^1 1 - \frac{1}{1+x^2}\,dx - \int_0^1 \pi x^2\,dx$$

$$= 2\pi - \frac{\pi}{4}2\pi - \frac{\pi}{3} = \frac{5}{3}\pi - \frac{\pi^2}{2}$$

1000.10.1.16

$-3a\cos^2 t\sin t$

$3a\sin^2 t\cos t$

16. 设星形线的方程为 $\begin{cases} x = a\cos^3 t, \\ y = a\sin^3 t \end{cases}$ $(a > 0)$，求：

(1) 它所围成的面积；

(2) 它的弧长；

(3) 它绕 x 轴旋转一周围成的旋转体的体积和表面积.

(1)

$$S = 4\int_0^{\frac{\pi}{2}} a\sin^3 t \cdot 3a\cos^2 t\,(-\sin t)\,dt$$

$$= -12a^2\int_0^{\frac{\pi}{2}} \sin^4 t\,(1-\sin^2 t)\,dt$$

$$= -12a^2\left(\frac{5}{6}-1\right)\frac{3}{4}\,\frac{1}{2}\,\frac{\pi}{2}$$

$$= 2a^2\,\frac{3}{4}\,\frac{1}{2}\,\frac{\pi}{2}$$

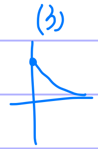
$$= \frac{3}{8}\pi a^2$$

(2) $$l = 4\int_0^{\frac{\pi}{2}} 3a\sqrt{\cos^4 t\sin^2 t + \sin^4 t\cos^2 t}\;dt$$

$$= 6a\int_0^{\frac{\pi}{2}} \sin 2t \cdot dt$$

$$= -3a\cos 2t\Big|_0^{\frac{\pi}{2}} = 6a$$

(3) $$V = 2\int_{\frac{\pi}{2}}^{0} \pi a^2\sin^6 t \cdot (-3a)\sin t\cos^2 t\;dt$$



$$= -6\pi a^3\int_0^{\frac{\pi}{2}} \sin^7 t\,(1-\sin^2 t)\,dt$$

$$= 6\pi a^3\left(1-\frac{8}{9}\right)\frac{6}{7}\,\frac{4}{5}\,\frac{2}{3}\cdot 1$$

$$= \frac{32}{105}\pi a^3$$

$$S = \int_0^{\frac{\pi}{2}} 6a \sin t \cos t \cdot 2\pi \cdot a \sin^3 t \, dt$$

$$= 12a^2\pi \int_0^{\frac{\pi}{2}} \cos t \, (1 - \cos^2 t)^2 \, dt \qquad \cos t \left[1 + \cos^4 t - 2\cos^2 t \right)$$

$$= 12a^2\pi \int_0^{\frac{\pi}{2}} \cos t + \cos^5 t - 2\cos^3 t \, dt$$

$$= 12a^2\pi \left(1 + \frac{4}{5}\frac{2}{3} - 2\frac{2}{3} \right) = \frac{12a^2\pi}{5}$$